

Free energy alchemical calculation of chemical potential in a dense Lennard-Jones fluid

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1 Introduction

The chemical potential is a central thermodynamic quantity which rules the behaviour of phase equilibria, transport, or particle exchange, among others. The ability to reliably estimate this quantity in a computationally efficient manner enables ab initio determination of phase diagrams, coexistence properties, and scalable exploration of the chemical space, which are essential for understanding and modelling complex molecular systems.

Dense Lennard-Jones fluids provide an ideal test case for such methods because they encapsulate the basic physical interactions of short-range repulsion and long-range attraction while remaining simple enough to allow the performance of controlled numerical experiments and comparison with high-quality reference data (as for example [1]). Moreover, the high densities considered here pose a challenge for standard insertion-based techniques, as successful insertions become increasingly rare in those regimes. Thus, Lennard-Jones systems of increasing density provide an incredibly relevant benchmark for developing, validating and assessing computational approaches to chemical potential estimation.

2 Methods

This project compares the performance of 5 of the most well-known techniques for free energy differences computation - Widom’s method, Thermodynamic Integration (TI), Free Energy Perturbations (FEP), Bennett’s Acceptance Ratio (BAR) and Multistate Bennett Acceptance Ratio (MBAR) - on the task of estimating the external chemical potential of a Lennard-Jones system at different values of the reduced temperature and density (T^*, ρ^*) that range from the gas state to a high-density fluid. As justified in some more detail in 2.3, this project uses a Langevin thermostat with the BAOAB splitting scheme as its integrator. The system is placed in a cubic box with periodic boundary conditions. Minimum image convention is used but no cut-off radius is specified, which means that the effective cut-off radius for the Lennard-Jones interactions is half of the length of the box. Code structure and design choices are explained in detail in 3.2, but it is important to remark that the code for this project follows a modular approach which means it would be easy to implement a different integrator and run all the functionalities with the new integrator instead. This project also has error analysis techniques

in place to estimate the uncertainty of the free energy estimators, as well as tail corrections to account for the finite cut-off radius in both external chemical potential and internal energy calculations.

2.1 Chemical Potential

The chemical potential of a system is defined as the energy change induced by the introduction of a new particle in the system, i.e.,

$$\mu = \frac{\partial G}{\partial N}_{|PTN} = \frac{\partial F}{\partial N}_{|VTN}$$

as described for example in [2, Eq. 7.2.1].

From now on, we will work with an NVT (canonical) ensemble. Thus, for sufficiently large N , we have that $\mu = F_{N+1} - F_N$. Let $H(p, q)$ be the Hamiltonian of our system of N particles, and suppose that we can write $H(p, q; \lambda)$ such that $H(p, q; 0)$ corresponds to the Hamiltonian of our N -particle system and $H(p, q; 1)$ corresponds to the Hamiltonian of our $(N+1)$ -particle system. Then, $F_N = F(\lambda = 1)$, but $F(\lambda = 0) \neq F_N$ because $F(\lambda = 0)$ is considering an extra, non-interacting particle. This difference satisfies $F(\lambda = 0) - F_N = -k_B T \ln \frac{(V/\Lambda^d)}{N+1}$. Thus, we can write $\mu = (F_{N+1} - F(\lambda = 0)) - (F(\lambda = 0) - F_N)$. We can rewrite $(F_{N+1} - F(\lambda = 0)) = F(\lambda = 1) - F(\lambda = 0)$, and we call this term the excess chemical potential μ_{ex} (intuitively, as it is the chemical potential introduced by the "non-ideality" of the added particle). We thus arrive at

$$\mu = \mu_{ex} - k_B T \ln \frac{(V/\Lambda^d)}{N+1}$$

where $\Lambda = h/\sqrt{2\pi m k_B T}$.

2.2 Free Energy Methods

We have effectively reduced the problem of calculating the chemical potential to a free energy difference calculation between two states of the Hamiltonian $H(p, q; \lambda)$. For the rest of this manuscript, we will assume a Hamiltonian with the structure

$$H(p, q; \lambda) = p^t M p + U(q, \lambda)$$

where M is a diagonal matrix with the weights of each of the particles of the system and U denotes the potential energy of the system, which we assume can be written as (abusing the notation):

$$U(q; \lambda) = \sum_{1 < i < j \leq N+1} V(q_i - q_j) + \lambda \sum_{i=2}^{N+1} V_\lambda(q_1 - q_i, \lambda)$$

Note that the inserted particle is given index 1 here to ensure consistency with Julia's 1-indexing, which is the chosen programming language for this project. From now on, let $\Delta F := F(\lambda = 1) - F(\lambda = 0)$, and let $\Delta F(\lambda) := F(\lambda) - F(\lambda = 0)$.

2.2.1 Thermodynamic Integration

The first option to compute the free energy difference encoded by μ_{ex} is to use thermodynamic integration. A foundational citation for this method can be attributed to [3]. The procedure, applied to our case, is the following: define the generalized force $\mathcal{F}(q, \lambda) = -\frac{\partial}{\partial \lambda} U(q; \lambda)$. Then, we have $\langle \mathcal{F} \rangle = -\frac{\partial}{\partial \lambda} F(\lambda)$. Thus, we can compute:

$$\Delta F = F(\lambda_1) - F(\lambda_0) = - \int_{\lambda_0}^{\lambda_1} \langle \mathcal{F}(q, \lambda) \rangle_{\lambda} d\lambda$$

2.2.2 Free Energy Perturbation

The second deeply established method this project considers is Free Energy Perturbations. A foundational citation for this method can be attributed to [4]. The method leverages the equality

$$e^{-(F_{\lambda'} - F_{\lambda})} = \frac{Z_{\lambda'}}{Z_{\lambda}} = \frac{\int e^{-(U_{\lambda'} - U_{\lambda})} e^{-U_{\lambda}} dq}{\int e^{-U_{\lambda}} dq} = \langle e^{-(U_{\lambda'} - U_{\lambda})} \rangle_{\lambda}$$

However, FEP is known to be very sensitive to the overlap between distributions at λ and λ' [5].

2.2.3 Bennet's Acceptance Ratio

Bennet's Acceptance Ratio (BAR) was introduced in [6], and it is an optimal estimator for the free energy between two states characterized by λ_1 and λ_2 . BAR states that the optimal estimation of the free energy difference between the two states must satisfy the following non-linear equation:

$$\sum_{i=1}^{N_A} \frac{1}{1 + \frac{N_1}{N_2} \exp[(\Delta U_{12}(q_i) - \Delta F)/k_B T]} = \sum_{j=1}^{N_B} \frac{1}{1 + \frac{N_2}{N_1} \exp[\beta(\Delta U_{21}(q_j) + \Delta F)/k_B T]}.$$

where N_1 is the number of samples taken from state 1 and N_2 is the number of samples taken from state 2.

Furthermore, we can derive the asymptotic variance of the BAR estimator, as done for example in [7]. The resulting expression is

$$\frac{1}{(k_B T)^2} \text{Var}(\Delta F) = \left[\sum_{i=1}^{N_1} \frac{1}{2 + 2 \cosh\left(\frac{\Delta U_{12}(q_i) - \Delta F}{k_B T} + \ln \frac{N_2}{N_1}\right)} + \sum_{j=1}^{N_2} \frac{1}{2 + 2 \cosh\left(\frac{\Delta U_{21}(q_j) + \Delta F}{k_B T} - \ln \frac{N_2}{N_1}\right)} \right]^{-1} - \frac{N_1 + N_2}{N_1 N_2}.$$

2.2.4 Multistate Bennet's Acceptance Ratio

Multistate Bennet's Acceptance Ratio (MBAR) was introduced in [7], and it is essentially the extension of BAR to the case where we have measurements from multiple states characterized by λ . It is traditionally defined using the reduced free energies and potentials, $u := \frac{1}{k_B T} U$ and $f := \frac{1}{k_B T} F$. MBAR requires solving a system of non-linear equations which would have an infinite number of solutions if we do not set the value of one of the free energies involved. For excess chemical potential estimation, we always choose $f_{\lambda=0} = 0$, as we are only interested in the free energy difference between the cases $\lambda = 0$ and $\lambda = 1$.

MBAR's system of equations for K different λ -states was first defined as

$$\hat{f}_k = -\ln \sum_{j=1}^K \sum_{n=1}^{N_j} \frac{\exp[-u_k(q_{j,n})]}{\sum_{m=1}^K N_m \exp[\hat{f}_m - u_m(q_{j,n})]}$$

where $N_1, \dots, N_K$ denote the number of samples available from each state.

However, we can observe that in this formula it does not matter which sample comes from each state inside the main summation. Thus, we take $N = \sum_{i=1}^K N_i$ and write:

$$\hat{f}_k = -\ln \sum_{n=1}^N \frac{\exp[-u_k(q_n)]}{\sum_{m=1}^K N_m \exp[\hat{f}_m - u_m(q_n)]}$$

For convenience, we define the weight matrix $W \in \mathbb{R}^{N \times K}$ as

$$W_{nk} = \frac{N_k \exp[\hat{f}_k - u_k(q_n)]}{\sum_{m=1}^K N_m \exp[\hat{f}_m - u_m(q_n)]}$$

Then, we can write the asymptotic covariance matrix as

$$\Theta = W^t (I_N - W \text{diag}(N_1, \dots, N_K) W^t)^+ W$$

where $+$ denotes a pseudo-inverse operation, as $I_N - W \text{diag}(N_1, \dots, N_K) W^t$ will be rank-deficient.

To implement MBAR, it is convenient to note that if we define $g_k(\hat{f}) = \hat{f}_k + \ln \sum_{n=1}^N \frac{\exp[-u_k(q_n)]}{\sum_{m=1}^K N_m \exp[\hat{f}_m - u_m(q_n)]}$, where we try to find the vector $\hat{f}$ that solves the system of non-linear equations $g(\hat{f}) = \mathbf{0}$, the Jacobian is given by $J_{\hat{f}} g = I - C$ where $C \in \mathbb{R}^K$ is defined as

$$C_{kl} = \sum_{n=1}^N \frac{N_l \exp[\hat{f}_l - u_l(q_n)]}{\sum_{m=1}^K N_m \exp[\hat{f}_m - u_m(q_n)]} \cdot \frac{\exp[\hat{f}_k - u_k(q_n)]}{\sum_{m=1}^K N_m \exp[\hat{f}_m - u_m(q_n)]}$$

and is commonly called the overlap matrix. Observe that $C = \text{diag}(N_1, \dots, N_K)^{-1} W^T W$.

2.2.5 Widom's Method

The last method for the computation of μ_{ex} considered by this project is Widom's method [8]. Widom's method has been included because it provides a good illustration of the challenges of

measuring μ_{ex} for increasingly dense systems. It takes

$$\mu_{ex} = -k_B T \int dq_{1:N+1} \langle e^{-\Delta U/k_B T} \rangle_N$$

where $\Delta U = U(N+1) - U(N)$. Observe that Widom's method can be compared to a free energy perturbation between the states $\lambda = 0$ and $\lambda = 1$, and thus requires a good overlap between the distributions for the completely non-interacting inserted particle and the fully interacting inserted particle. This can only hold true at low densities, which is why we expect Widom's method to fail at high densities.

2.3 Molecular Dynamics Simulations

In order to be able to compute μ_{ex} through any of the methods we have introduced, we need to be able to run molecular dynamics (MD) simulations on the canonical ensemble. Thus, the standard Verlet integrator is not applicable to our problem. Instead, we will use a Langevin integrator with the BAOAB scheme. Langevin dynamics introduce a stochastic perturbation of the Hamiltonian dynamics that, while generating trajectories that may not match the "physical" Hamiltonian trajectories, guarantee ergodicity and convergence to the stationary probability distribution function of the canonical ensemble [9]. In particular, this means that the Langevin update acts as a thermostat. Further, Langevin dynamics with the BAOAB splitting have second-order accuracy and are robust to the choice of longer timesteps [10]. One step of Langevin dynamics with BAOAB splitting for a given Hamiltonian $H = \frac{1}{2}p^t M^{-1}p + U(q)$ is given by:

$$\begin{aligned} p_{n+1/4} &= p_n - \frac{\Delta t}{2} \nabla U(q_n) \\ q_{n+1/2} &= q_n + \frac{\Delta t}{2} M^{-1} p_{n+1/4} \\ p_{n+3/4} &= e^{-\gamma \Delta t} p_{n+1/4} + \sqrt{k_B T M (1 - e^{-2\gamma \Delta t})} z_n \\ q_{n+1} &= q_{n+1/2} + \frac{\Delta t}{2} M^{-1} p_{n+3/4} \\ p_{n+1} &= p_{n+3/4} - \frac{\Delta t}{2} \nabla U(q_{n+1}) \end{aligned}$$

where $z_n \sim \mathcal{N}(\mathbf{0}, I)$.

In this project, simulations are run in a cubic box of side L with periodic boundaries to simulate as if we had an infinite number of particles. This eliminates edge effects and will allow us to compute the desired thermodynamic properties for dense Lennard Jones fluids as if every particle was in the "bulk" of a liquid. We implement minimum image convention, which means that every particle can only "feel" the closest of the infinite reflections of every other particle and does not interact with any of its own infinite reflections. This corresponds to setting a cut-off radius of $r_c = \frac{L}{2}$.

2.4 Statistical Methods for MD

While we know that Langevin dynamics have the ergodicity property, i.e., once equilibrium is reached we can measure the expected values of thermodynamic quantities by averaging over

the sum of the values at each visited configuration during the MD simulation, these samples are not independent. While this does not represent a problem for computing means, it does introduce a challenge for the computation of uncertainty estimates. The problem is that running these MD simulations is computationally heavy, so we cannot afford to measure the standard deviation (std) of our estimators by recomputing the MD simulations multiple time to get multiple different estimations that we can use to calculate the standard deviation. However, we have also mentioned that the samples we take can be very correlated, and this correlation would corrupt the estimation of the std, which requires independent samples.

2.4.1 Integrated Autocorrelation Time

To solve this issue, we first need to introduce the integrated autocorrelation time τ , which measures after how many simulation steps does a sample become effectively independent of its previous values. The integrated autocorrelation time is directly related to statistical inefficiency $g = 1 + 2\tau$, which is the factor by which the variance of a series of correlated samples is underestimated [11]. We can compute τ as

$$\tau = \frac{1}{2} + \sum_{k=1}^{\infty} \frac{C(k)}{C(0)}$$

where $C(k) = \sum_{i=1}^{l-k} \frac{(v_{i+k} - \mu_v)(v_i - \mu_v)}{l-k}$ is the estimator of the correlation between the vector of samples v and itself but lagged by k steps [2]. Thus, $C(0)$ is just the estimator for the variance of v . Please allow me to get ahead of ourselves and indicate that what we observed in practice is that this estimator has a very high variance and can sometimes return unphysical negative predictions that are difficult to deal with. Thus, we introduce here Sokal's proposed approach, which is used in software packages such as `emcee` [12].

Sokal proposes to take a biased estimator of the correlation times $C(k)$ by dividing by l instead of $l - k$ in the formula above. This choice reduces the variance and guarantees positive results. Further, Sokal's criterion gives a heuristic to stop the summation for the computation of τ . The rule is to stop the summation once the lag k is significantly larger than the estimated τ , i.e., to stop when $k \geq c \cdot \tau$ where typically $c \approx 4$ or 5 [13].

2.4.2 Block Averaging

Having defined the integrated autocorrelation time and given tools to compute it, we are ready to tackle the challenge of estimating uncertainty with correlated samples. Block averaging consists on dividing the data on n_B blocks of size B which will act as a sort of "independent statistical units" if $B \gg \tau$. We then compute the mean of the samples for each of these blocks, which will be independent, and thus compute the uncertainty as the standard error of the mean of the block means:

$$\mu_i = \frac{1}{B} \sum_{j=(i-1)B+1}^{iB} v_j \quad \text{for } i \in \{1, \dots, n_B\}$$

$$\sigma_v^2 \approx \frac{1}{n_b(n_b - 1)} \sum_{i=1}^{n_b} \left[\mu_i - \left(\frac{1}{n_b} \sum_{i=1}^{n_b} \mu_i \right) \right]^2$$

where v is our vector of samples [2].

2.4.3 Bootstrapping

Block-averaging has one main limitation, and that is that it cannot deal with non-linear estimators without having to approximate the propagation of the uncertainty through the non-linear function. Bootstrapping addresses this limitation by treating each block as an independent statistical object and then resampling "synthetic" datasets by sampling blocks with replacement. By applying the estimator to each of these resampled datasets we can estimate the std of our estimator as its standard deviation on the discrete distribution given by our resampled datasets [2]. Note that increasing the number of resampled datasets does not result in an underestimation of the std, it just yields a less noisy estimator for it. However, the std will actually be underestimated if our number of effective samples ($N_{eff} = N/g$) is too low. We can formalize this procedure as:

- Resample datasets $D_1, \dots, D_{n_{boot}}$ by taking n_B blocks from the original blocks with replacement.
- Given the estimator $f(D)$, compute

$$\sigma_{boot}^2 = \frac{1}{n_{boot} - 1} \sum_{i=1}^{n_{boot}} \left[f(D_i) - \left(\frac{1}{n_{boot}} \sum_{i=1}^{n_{boot}} f(D_i) \right) \right]^2$$

2.5 Lennard-Jones Potential

The Lennard-Jones potential was introduced in [14], and it models the potential between two particles as

$$V_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

where ϵ and σ represent the well depth and the collision diameter, respectively.

For the computation of μ_{ex} it is necessary to introduce a softened Lennard-Jones potential which introduces a dependency on λ , namely $V_{LJ,\lambda}(r; \lambda) = 4\epsilon(A^{-2} - A^{-1})$, where $A = \frac{1}{2}(1 - \lambda) + (\frac{r}{\sigma})^6$.

Recall that we will be working with an effective cut off radius of $\frac{L}{2}$. Thus, we will introduce tail corrections using the assumption that $\rho(r) \approx 1$ for all $r > \frac{L}{2}$ for those reduced pressure - reduced temperature points in which we are in the fluid state. The tail correction under such assumptions for a generic potential is $U_{tail} = \frac{N}{2} 2\pi\rho \int_{r_c}^{\infty} U(r) r^2 dr$, so substituting for the Lennard-Jones potential we get

$$U_{tail}^{LJ} = \frac{8\pi\rho\epsilon\sigma^3}{3} \left[\frac{1}{3} \left(\frac{\sigma}{r_c} \right)^9 - \left(\frac{\sigma}{r_c} \right)^3 \right].$$

Now, we can simply obtain μ_{ex}^{tail} leveraging the fact that $\mu_{ex}^{tail} = \frac{2}{N} U_{tail}$.

Finally, we introduce the reduced, dimensionless units for the Lennard-Jones potential. These are defined by choosing $\sigma = 1$, $\epsilon = 1$, $m = 1$. Then, physical quantities are rescaled as

$$r^* = \frac{r}{\sigma}, \quad \rho^* = \rho\sigma^3, \quad T^* = \frac{k_B T}{\epsilon}, \quad U^* = \frac{U}{\epsilon}.$$

In reduced units, the LJ potential becomes $U_{LJ}^*(r^*) = 4[(r^*)^{-12} - (r^*)^{-6}]$.

3 Implementation details

3.1 Experimental setup

We performed experiments at 4 different reduced pressure-reduced temperature points, which were chosen to showcase both the behaviour of μ_{ex} as a function of the density and the performance of each of the considered methods. The chosen points were

$$(T^*, \rho^*) = (1.0, 0.8), (1.0, 0.9), (1.25, 0.45), (1.5, 0.1)$$

These points were selected by consulting a NIST reference dataset [1]. The point $(T^*, \rho^*) = (1.5, 0.1)$ corresponds to a supercritical gas state (the critical temperature is $T^* = 1.31$), this is, an LJ system in a very low density gas state. The point $(T^*, \rho^*) = (1.25, 0.45)$ corresponds to a system that has already undergone a transition to liquid phase, but still has a relatively low density. Finally $(T^*, \rho^*) = (1.0, 0.8), (1.0, 0.9)$ correspond to a very dense LJ fluid, with the triple point density being $\rho^* \approx 0.85$.

What we expect is that, for sufficiently low densities, there is plenty of space for the inserted particle to sit in the attraction well of the other particles and thus we will obtain negative values of μ_{ex} . However, from a certain density on we expect the system to become too crowded, so μ_{ex} should attain positive values which will grow exponentially if we kept augmenting the pressure. In terms of shape, for densities that are not extremely high (like the ones we will be working with) we expect something resembling a negative parabola, were the inserted particle first "clashes" with the pre-existing particles (yielding a positive increment of the free energy) but then finds its own "well" inside the system, weighing down the total free energy difference.

We settled on $\delta t = 0.002$ in reduced units for the simulations at $(T^*, \rho^*) = (1.0, 0.8), (1.0, 0.9)$ because this kept manageable autocorrelation times while still "safe" at these higher densities in which repulsions are much more frequent. Further, we chose a λ schedule with 20 steps for TI and BAR/MBAR and a λ schedule with 50 steps for FEP. For the simulations at $(T^*, \rho^*) = (1.25, 0.45), (1.5, 0.1)$ we moved to $\delta t = 0.005$ (in reduced units) because the autocorrelation times (measured in steps) in these low density regime were otherwise too high to obtain a relevant number of effective samples. Further, we chose a λ schedule of 50 steps for TI and BAR/MBAR and double that for FEP at $\rho^* = 0.8$, and we only report results at $\rho^* = 0.9$ for TI and BAR/MBAR -for which we use 100 λ steps- because we observed FEP to become unstable at these densities. Last but not least, we also settled on $\gamma = 1.0$ because it was observed to produce nice results.

Particles are initialized at uniform positions on a cubic box with length such that the desired density is obtained. We performed simulations with an initial equilibration of 50000 steps to get rid of the initial unphysical state, an equilibration of 15000 steps between different λ values and a production run of 20000 steps for each λ value.

We used block averaging for TI and bootstrapping for FEP (in the case of FEP, we can perform bootstrapping on the block means of the samples $e^{-\Delta U/k_B T}$, and this is exactly the method we follow, taking `n_boot` = 1000). We also performed a histogram analysis on Widom method to illustrate its shortcomings. We report both the bootstrapped (with `n_boot` = 100 to retain computational efficiency) and the theoretical asymptotic std for BAR. We also implemented a function to compute the theoretical asymptotic std of MBAR. However, we do not use it in our experiments because it requires taking a pseudo-inverse which we observed to be very

ill-conditioned when only subsampled (effectively independent) data is fed to it, but becomes too slow if we try to use all our samples (doing this would further be wrong, as the formulas for MBAR covariance assume independent samples).

3.2 Method implementation and code structure

Even though a conscious choice to work in reduced units for the sake of generality was made in this project’s implementation, the code provide functions to automatically convert these into standard units. Further, the code can handle any Hamiltonian that can be written in the following form: $H(p, q; \lambda) = p^t M p + U(q) + U_\lambda(q; \lambda)$, where in our case all particles have the same mass (1 in reduced units) and thus $M = mI$. Nevertheless, the syntax is biased towards the case where $U_\lambda(q; \lambda) = \lambda \sum_{i=2}^{N+1} V(q_1 - q_i, \lambda)$, which is the form taken by the potential in our case. To speed up computations, the code is already equipped with explicit expressions for $\frac{dV}{dr}$, $\frac{d(\lambda V)}{d\lambda}$ and $\frac{d(\lambda V)}{dr}$ for $V = \text{LJ potential}$ and $V = \text{softened LJ potential}$. Please note that, because the code has been designed for computing the free energy difference of particle insertion, to simulate a standard system of N LJ particles we technically have to call our integrators with $\lambda = 1$. One could be tempted to set $\lambda = 0$, but that would yield a system with N particles in which one of them is completely non-interacting.

To keep generality, we will report only μ_{ext}^* . The ideal contribution to the final chemical potential requires the use of the Planck constant h (or alternatively, the reduced Planck constant, but this would require the dimensionfull values of σ , ε and m). However, the code includes a function for the computation of the ideal chemical potential to the chemical potential which is ready to use.

We provide separate functions and integrators for running MD and collecting samples for each of the methods we considered to maximize performance. We provide data structs to store the relevant sampled values for TI, FEP and BAR/MBAR. This allows us to separate the process of running MD simulations from the evaluation the predicted free energy differences without having to store all sampled positions of the particles. There is one possible inefficiency here, and that is that our current approach currently involves running three different production simulations to get the data for TI, FEP and BAR/MBAR. However, this is a conscious choice, as we wanted to retain the possibility to run simulations with different parameters for each of the methods. Note that we obtain the data for BAR and MBAR simultaneously because for each simulation at λ we can compute the evaluation of $U_{\tilde{\lambda}}(q; \tilde{\lambda})$ for all $\tilde{\lambda}$ in a vectorized manner that does not incur in a large computational overhead.

Widom’s method is implemented in practice by simulating with N particles and taking a given number of trial insertions every certain number of timesteps. We take 500 insertions every 1000 steps to avoid statistical inefficiency, because we have observed that the integrated autocorrelation time can sometimes take values of over 400 steps. We repeat this 1000 times for a total of 500000 trial insertions.

Thermodynamic integration is implemented using the trapezoid method. Thus, when we apply block-averaging, the predicted std propagation is ruled by the following result: if we take the random variable $Y = \sum_{i=1}^k w_i X_i$ where X_i are all independent random variables, then $Var(Y) = \sum_{i=1}^k w_i^2 Var(X_i)$ (by the standard variance rules). Observe that integrating with the trapezoid method is equivalent to taking a weighted sum of the realization of independent random variables (the expected value estimators) at each value of λ . Thus, we get (assuming

a uniform λ schedule of $K + 1$ values for convenience, which is what we have chosen):

$$\text{std}(\Delta F_{TI}) = \left(\frac{1}{2}\delta\lambda\right)^2\sigma_0^2 + \sum_{i=2}^K \delta\lambda^2\sigma_{\lambda_i}^2 + \left(\frac{1}{2}\delta\lambda\right)^2\sigma_1^2$$

Note that we can also use this formula to calculate $\text{std}(\Delta F_{TI}(\lambda_i))$ by simply taking λ_i as the last point of the summation (thus its weight is half). This result assumes that our expected value estimators across λ windows are independent. Nevertheless, this is a reasonable assumption because we equilibrate our system every time we change λ before sampling. Future work would involve testing this assumption and computing the std of the TI method by bootstrapping the TI free energy estimator as a whole.

The propagation of the predicted std by the bootstrapping method in the case of FEP draws from the same result, but in this case we have the simple increment

$$\text{std}(\Delta F(\lambda_{i+1})_{FEP}) = \sqrt{\text{std}(\Delta F(\lambda_i)_{FEP})^2 + \sigma_{\lambda_{i+1}}^2}$$

BAR is straightforward to implement, as it just requires finding the zero of a non-linear function. In order to do this, I allowed myself to make use of the function `find_zero` in the Julia module `Roots`. However, to respect the requirement that all code is produced by ourselves, I implemented MBAR’s solver from scratch with the following strategy: we first compute 50 iterations of Picard’s method to warm start the algorithm and then perform iterations of Newton’s method until convergence. Note that the matrix of reduced energies for all λ states for the samples from each λ state is constructed by vertically concatenating the matrices of the reduced energies for each λ value for the samples from a fixed λ state. The Jacobian matrix is constructed as described in the theory section. As suggested in [7], for numerical stability we always take $\ln \sum_{i=1}^l e^{x_i} = m + \ln \sum_{i=1}^l e^{x_i - m}$ where $m = \max_{i \in \{1, \dots, l\}} x_i$.

All the code to perform bootstrapping on BAR and MBAR is in place but we only use it for BAR because it is quite computationally expensive for MBAR. Initially, the implementation defaulted to subsampling, as this is consistent with BAR and MBAR theoretical formulation - which assumes independent samples. However, subsampling impedes MBAR convergence (we tested for up to 10000 iterations and convergence was not observed in the subsampled regime). Nevertheless, it is common practice to use BAR and MBAR with correlated samples, as it has been observed that they still produce correct predictions [15]. Thus, we use all samples to estimate the free energy difference and then subsample to obtain the theoretical prediction for the std.

4 Experiments

4.1 Verification

We verified that the function for the LJ potential and the BAOAB integrator were correct by manually computing one BAOAB step for the case where we have two particles at positions $(\pm 1, 0, 0)$ and the inserted particle is at $(0, 0, 0)$.

We verified that the Langevin dynamics are correctly thermostating the system by checking the evolution of the kinetic temperature for the system at the 4 temperature-density points we have studied (see figure 4).

We use $N=256$ because we tried $N=500$ and observed almost exactly the same potential energy per particle (-5.522 for $\rho = 0.8$ and $T = 1.0$) than at $N=256$ (-5.516 for the same values), so we are in the thermodynamic limit (or very close to it). This matches the theoretical prediction obtained by computing the energy of the system by numerically integrating the theoretical formula

$$U = \frac{N}{2} 2\pi\rho \left(\int_0^{r_c} U(r) g(r) r^2 dr + \int_{r_c}^{\infty} U(r) r^2 dr \right)$$

using the computed values for the rdf function and the expression for U_{tail} . This yields -5.4223 for $N = 256$, which confirms that our implementation is sound. The plot of the rdf function for a standard LJ system at for a system of N particles with $\lambda = 0.5$ for the inserted particle (which are indistinguishable because a single inserted particle which is behaving approximately like the others does not produce noticeable changes in the structure of the fluid) can be found in figure 1.

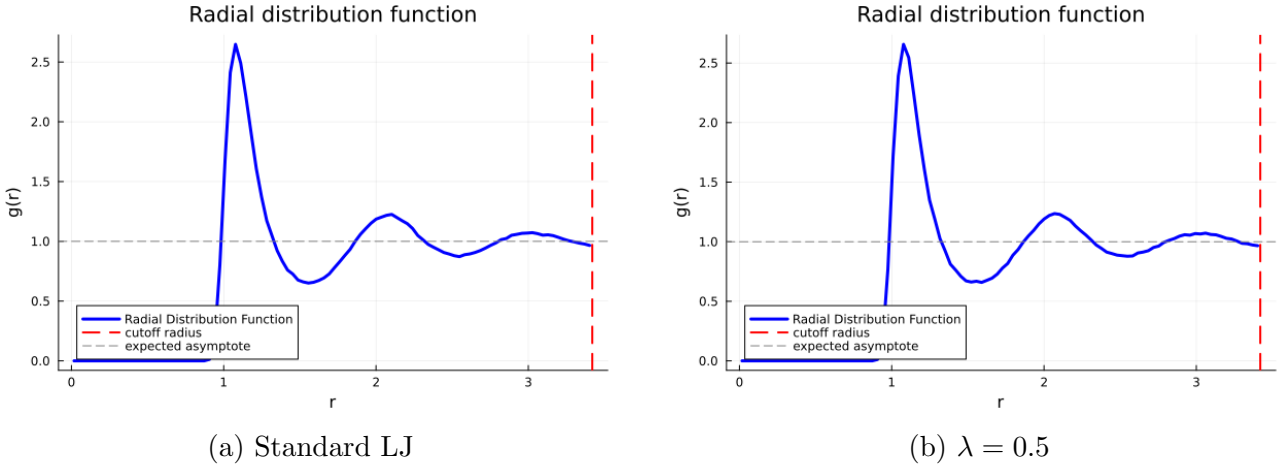


Figure 1: Rdf function for a standard LJ system and an LJ system with an inserted particle with interaction tuning $\lambda = 0.5$ at $(T^*, \rho^*) = (1.0, 0.8)$

4.2 Results and Discussion

First, we tried equilibrating a system with 256 particles at $\lambda = 0$ and then moving to $\lambda = 0.1$ without the softened core potential, and we observed that the free energy difference exploded. Thus, we report results using exclusively the softened core potential for the interactions of the inserted particle.

4.2.1 $T^* = 1.5, \rho^* = 0.1$

At this reduced temperature and pressure, the system is effectively in a very low-density gas state. This very low density means that the inserted particle can very easily sit in the attractive well of the rest of the particles of the system, so the free energy difference as a function of λ barely raises above zero when we "turn on" the interactions for the first time, and thus starts decreasing almost immediately (see figure 5). However, we only see a moderate negative slope because the particles can sit so far away from each other that the attraction forces between them are very small, yielding a final prediction for μ_{ex} of around -0.68 (see table 1).

The tail correction here is almost negligible because our cut-off radius is very large due to the box size being chosen to achieve the desired ρ^* . Unsurprisingly, the predicted experimental std is very small for TI and FEP (≈ 0.02), so the predictions approximately sit in the 1 std margin of one another except for the case of MBAR. This discrepancy is however easily explained when we consider BAR’s asymptotic std, which is 0.03549 and leaves MBAR’s prediction right next to the 1 std margin (the bootstrapped BAR std is approximately one third of the asymptotic value). This discrepancy is thus probably explained by the low number of effective samples - which is induced by the high autocorrelation times observed for λU_λ , which reach a maximum of 422 steps as a function of λ .

	TI	FEP	BAR	MBAR	Widom
μ_{ex}^*	-0.68034	-0.7029	-0.6869	-0.6445	-0.6922
$\mu_{\text{ex}}^{\text{tail}*}$	-0.002618	-0.002618	-0.002618	-0.002618	-0.002618

Table 1: Dimensionless excess chemical potential and tail corrections computed by different free-energy methods at $(T^*, \rho^*) = (1.5, 0.1)$

4.2.2 $T^* = 1.25$, $\rho^* = 0.45$

When we move to a reduced temperature of 1.25 and a reduced pressure of 0.35 we are simulating an LJ system already in the liquid state, but still in a relatively low density regime. This means that the initial repulsions are stronger, so the free energy difference as a function of λ raises above 0 before decreasing sharply because the inserted particle can still find a place in the attraction well of the remaining other particles of the system (see Figure 6). Even though this situation mimics what we already observed for the previous case, the difference lies in the fact that the particles now all sit much closer to one another (even if still in the attraction well) and thus we see that we obtain a significantly low prediction for μ_{ex}^* , of approximately -2.6 (see table 2)

TI, FEP have stds between 0.03 – 0.04. We observe that BAR’s asymptotic std is 0.08354, and the std predicted by bootstrapping is approximately half of that. Note that the predictions of FEP, BAR and MBAR all lie within the 1 (asymptotic) std margin. TI’s prediction is slightly off (but still within the 2 std margin), which might be explained by the relatively coarse λ schedule or the long time step of 0.005. Thus, for the remaining cases we use both finer λ schedules and a more conservative time step of 0.002 (in reduced units). It is interesting to observe that Widom’s method was still able to arrive at a correct estimate for this value of the density.

	TI	FEP	BAR	MBAR	Widom
μ_{ex}^*	-2.5481988	-2.6038	-2.6803	-2.6690	-2.62417
$\mu_{\text{ex}}^{\text{tail}*}$	-0.05301	-0.05301	-0.05301	-0.05301	-0.05301

Table 2: Dimensionless excess chemical potential and tail corrections computed by different free-energy methods at $(T^*, \rho^*) = (1.25, 0.45)$

4.2.3 $T^* = 1.0$, $\rho^* = 0.8$

In this regime, we already have a very dense LJ fluid, with ρ^* close to the triple point value. Thus, we expected to see the value of μ_{ex} to increase with respect the value we previously

obtained for the case $T^* = 1.25$, $\rho^* = 0.45$. That is exactly what we observe: the final free energy difference has now started to increase, taking a value of approximately -2.1 before the tail correction, which is now much more significant with a value of -0.1675 . It makes sense to have larger tail corrections as we increase the density of our system because the cut-off radius, which is just $L/2$, becomes smaller.

The results for each method are shown in table 3. An illustration of the free energy difference evolution as a function of λ can be found in figure 2. Regarding BAR’s uncertainty, note that once again we observe that the std predicted by bootstrapping (≈ 0.06) underestimates the asymptotic std (≈ 0.1) - which is exactly what we were expecting. Observe that the predictions of all MBAR lie within this uncertainty bound, but the predictions given by TI and FEP are 2 stds away. It is also interesting to observe that the std for TI and FEP closely matches the bootstrap std for BAR.

It is surprising to see Widom’s method retain a significant accuracy. This is probably due to the high number of trial insertions (500000) we are performing.

Finally, observe that the free energy difference as a function of λ predicted by FEP is still significantly ragged, which might be an indicator of poor overlap. Moving to the final case, we would need to once again increase the number of λ steps to obtain reliable results with FEP, which makes it significantly more expensive than running TI, or even running BAR/MBAR. Further, we observed that keeping this higher number of λ steps is essential for FEP’s performance, as it produced significantly wrong results for this case when we tried using only 50 λ steps instead of 100 (this is due to its need for a high overlap between the distributions). Thus, we only report results for TI, BAR, MBAR and Widom for the final case.

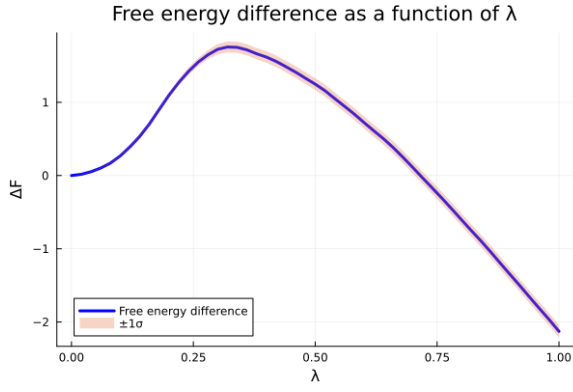
	TI	FEP	BAR	MBAR	Widom
μ_{ex}^*	-2.12803	-2.0929	-2.3428	-2.3296	-2.1560
$\mu_{\text{ex}}^{\text{tail}*}$	-0.1675	-0.1675	-0.1675	-0.1675	-0.1675

Table 3: Dimensionless excess chemical potential and tail corrections computed by different free-energy methods at $(T^*, \rho^*) = (1.0, 0.8)$

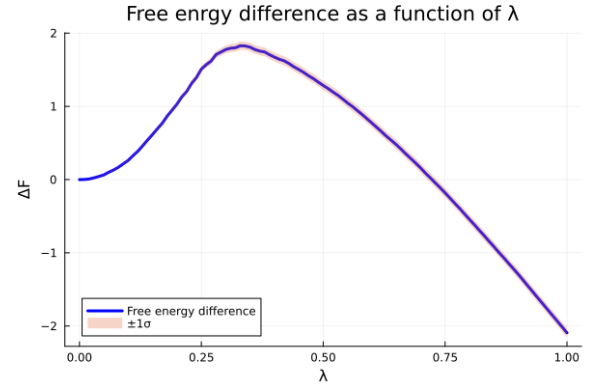
4.2.4 $T^* = 1.0$, $\rho^* = 0.9$

In this state, $\rho^* = 0.9$ is just above the triple point value, placing the system near the onset of the solid-liquid coexistence region. As a result, we observe an exponential leap in the predicted value of μ_{ex}^* wrt the case $T^* = 1.0$, $\rho = 0.8$, with μ_{ex}^* now being positive. In other words, the inserted particle is not able to find a pocket in the attraction well of the rest of the system’s particles after the initial repulsion when the interactions are turned on. This means that the free energy as a function of λ has a strong initial increase corresponding to a strong initial repulsion and then decreases but does not become negative. Nevertheless, the predictions made by our different methods are far off from each other, as showed in table 4

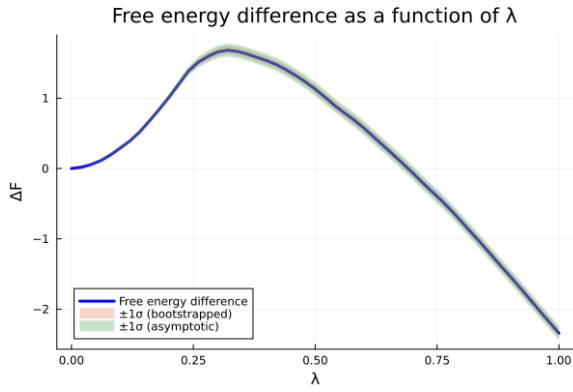
TI reports an std of ≈ 0.09 , which means that BAR and MBAR’s predictions lie within the 2 std range of the TI prediction but far from the 1 std margin. We only report BAR’s asymptotic std because bootstrapping becomes very computationally expensive, and surprisingly we observe that BAR’s asymptotic covariance (≈ 0.07) is now lower than TI’s experimental std (computed by block-averaging). This seems to indicate that TI would need a finer schedule or a smaller time step to produce a reliable estimate, as it is otherwise unusual for the std encoded in the



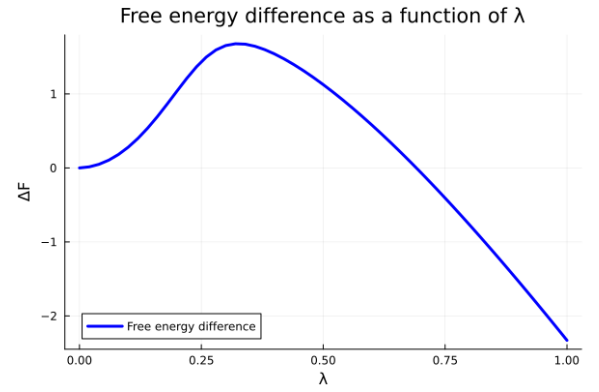
(a) TI



(b) FEP

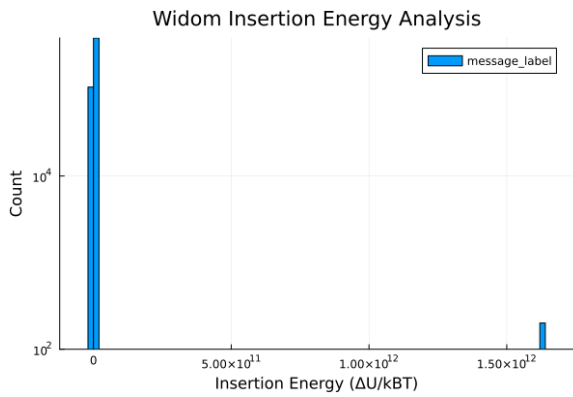


(c) BAR

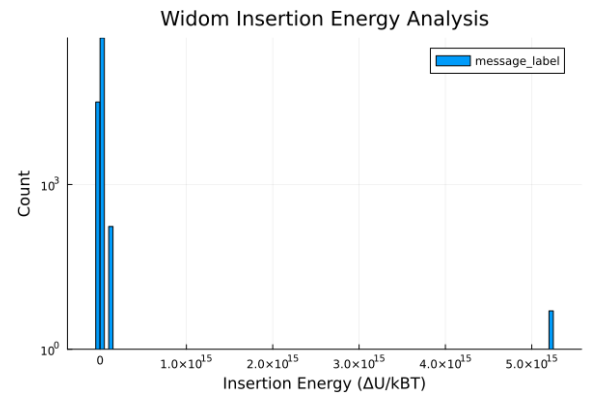


(d) MBAR

Figure 2: Free energy difference as a function of λ for TI, FEP, BAR and MBAR at $(T^*, \rho^*) = (1.0, 0.8)$



(a) $(T^*, \rho^*) = (1.5, 0.1)$



(b) $(T^*, \rho^*) = (1.0, 0.9)$

Figure 3: Comparison of the insertion energies for Widom's method at two different (T^*, ρ^*) points

finite data seen by block-averaging to be greater than the theoretical, asymptotic std for BAR. Another diagnosis that points in this direction is the fact that TI’s free energy difference as a function of λ has become ragged around the peak (see Figure 7), so a finer schedule only for those values of λ around which the peak is reached could be a plausible solution. BAR’s free energy difference as a function of λ is also slightly ragged, but in this case we can trust its prediction as it lies well within the 1 std range of MBAR’s solution, which is the only method capable of producing a perfectly smooth prediction of the free energy difference as a function of λ .

Finally, note that this state is completely inaccessible for Widom’s method, whose prediction is not even reconcilable with the others. This is because the density has now become so high that the probability of a successful insertion is extremely low. When we plot the histogram of the insertion energies, we observe that a significant portion of the insertion energies take astronomically large values (order $1e15$). Figure 3 illustrates this.

	TI	FEP	BAR	MBAR	Widom
μ_{ex}^*	0.4193	-	0.2247	0.2377	1.8247
$\mu_{\text{ex}}^{\text{tail}*}$	-0.2120	-0.2120	-0.2120	-0.2120	-0.2120

Table 4: Dimensionless excess chemical potential and tail corrections computed by different free-energy methods at $(T^*, \rho^*) = (1.0, 0.9)$

5 Limitations and Future Work

The main limitation of this project is the number of effective samples generated. For some of the longest correlation times observed (of around 500 steps), once we subsample every $2 \times 500 = 1000$ steps we are left with only 20 effective samples from our production run. This is enough to run some approximate estimation, but far from providing solid results. The reason why we did not take longer production runs, finer λ schedules, more equilibration steps between λ values, etc. is due to the computational cost one would need to incur in to take larger quantities of any of these values.

Thus, it would be interesting to adapt the existing codebase to be compatible with CUDA acceleration. Further, to facilitate its use in "real" computational tasks, the code should include the ability to control the cut-off radius and implement neighbour lists to translate this cut-off radius into an increase in computational efficiency.

Another unexplored direction was the evaluation of the performance of other relevant free energy estimators. The most renowned estimator to be left out is perhaps Jarzynski’s equality [16]. A small literature study convinced me not to use this method because it would be quite sensitive to the followed path. However, it would have still been interesting to consider the escorted Jarzynski’s equality [17], which can theoretically provide a solution to this sensitiveness. Lastly, it would be interesting to compare the predictions of these classical methods with a ML-based method such as FEAT [18].

Author disclaimer

The author acknowledges the use of LLMs for help coding the plotting functions involved in the project as well as for testing & refining his understanding of the implemented methods, and occasionally debugging when the situation was dire.

Code Availability

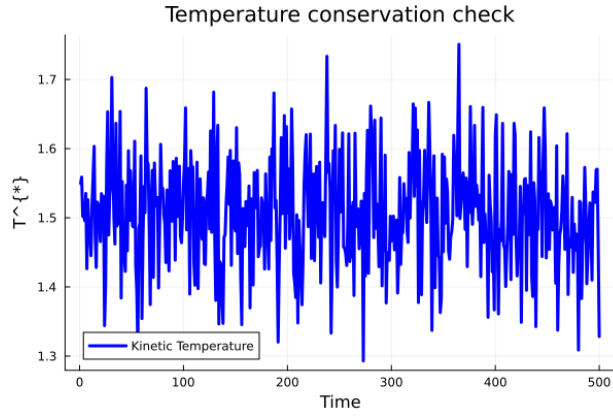
All code is available at <https://github.com/javier-gallardo-saenz/muFE-in-LJFluid> and implemented in the Julia language.

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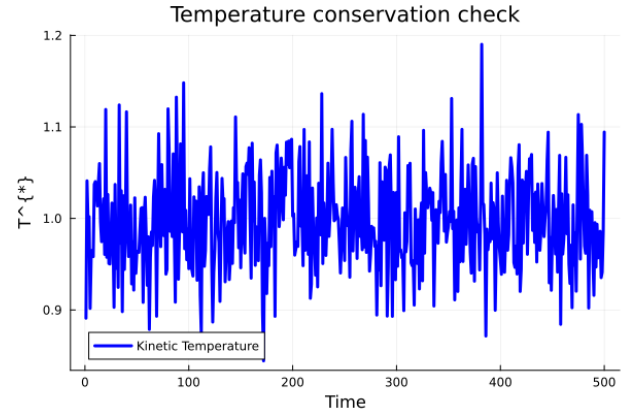
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Appendix

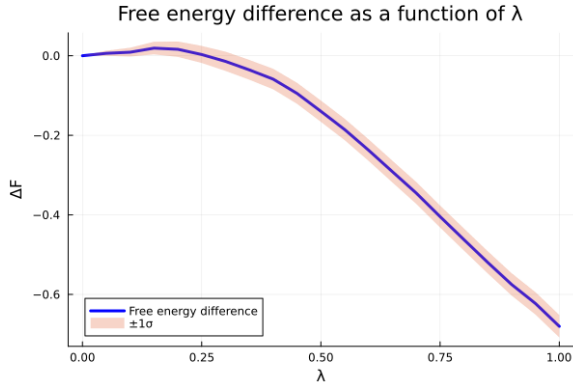


(a) $(T^*, \rho^*) = (1.5, 0.1)$

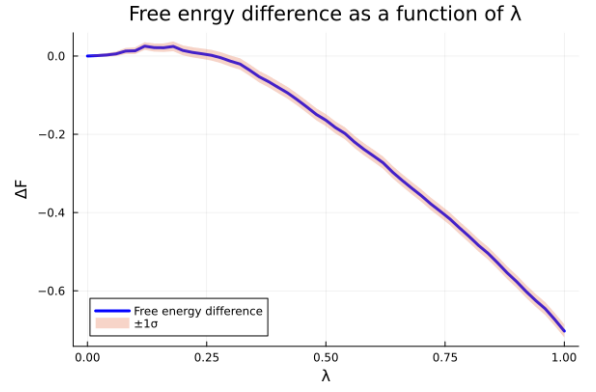


(b) $(T^*, \rho^*) = (1.0, 0.8)$

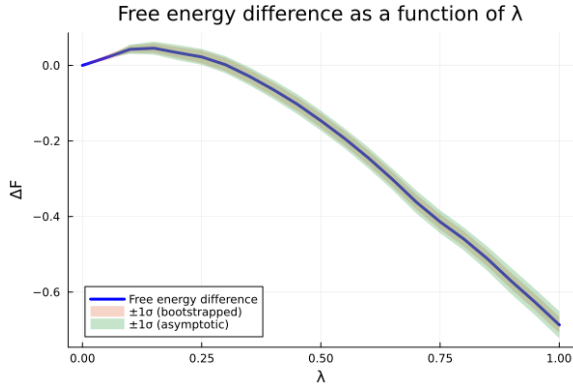
Figure 4: Temperature evolution along Langevin BAOAB dynamics of an LJ system at different states (T^*, ρ^*)



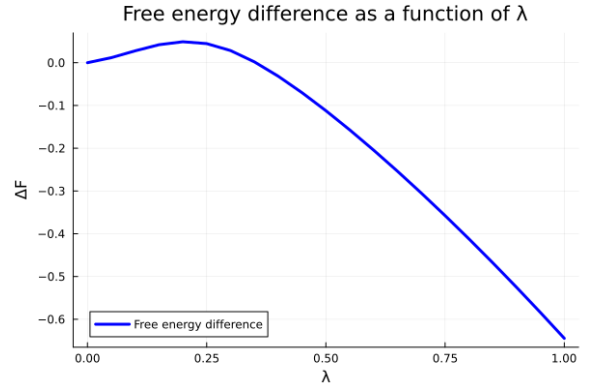
(a) TI



(b) FEP

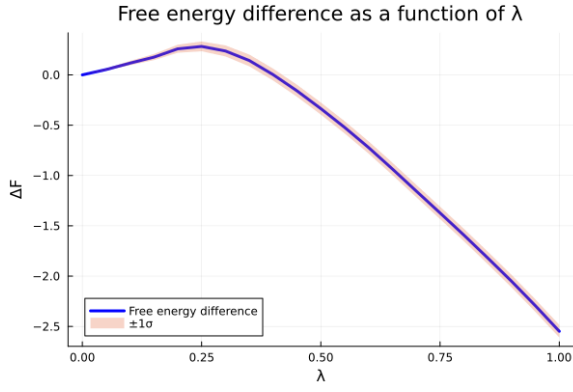


(c) BAR

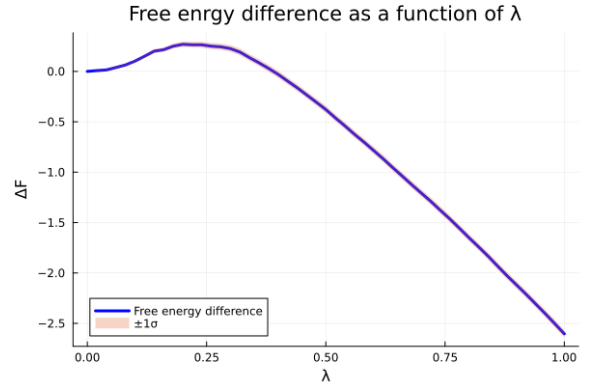


(d) MBAR

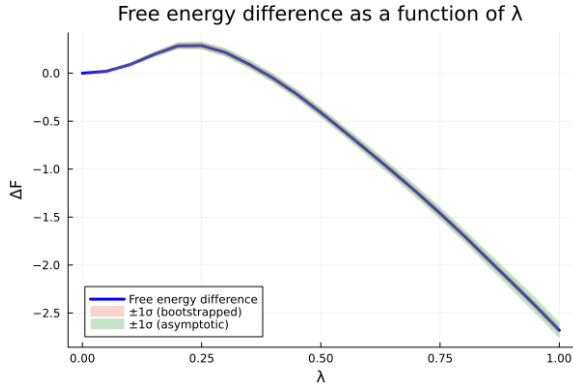
Figure 5: Free energy difference as a function of λ for TI, FEP, BAR and MBAR at $(T^*, \rho^*) = (1.5, 0.1)$



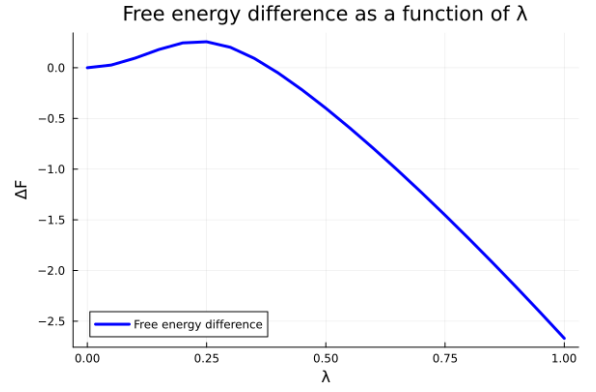
(a) TI



(b) FEP

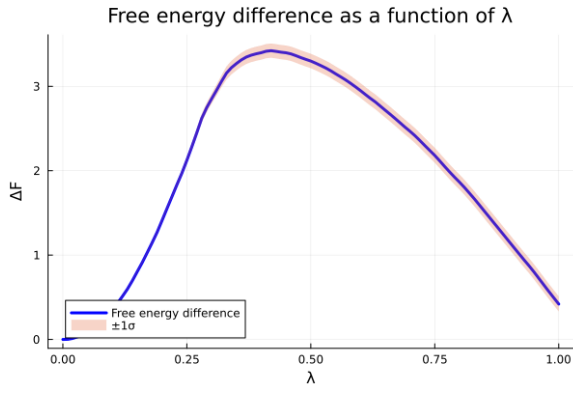


(c) BAR

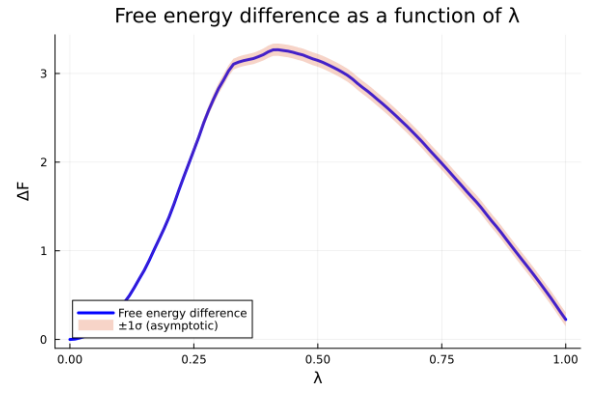


(d) MBAR

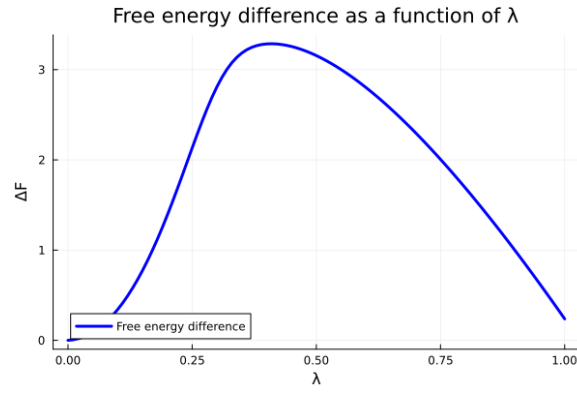
Figure 6: Free energy difference as a function of λ for TI, FEP, BAR and MBAR at $(T^*, \rho^*) = (1.25, 0.45)$



(a) TI



(b) BAR



(c) BAR

Figure 7: Free energy difference as a function of λ for TI, FEP, BAR and MBAR at $(T^*, \rho^*) = (1.0, 0.9)$